

Silvaco TCAD Tutorial and Simulation Guide for Semiconductor Device Modeling (Beginner Level)



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github.com/Neethu-Jaisan/Silvaco_TCAD

Introduction

Welcome to the **Silvaco TCAD Tutorial**, a beginner-level guide designed to help newcomers confidently navigate the world of Technology Computer-Aided Design. Rooted in the Silvaco TCAD ATLAS Tutorial video, this documentation will walk you step-by-step through mastering foundational process and device simulations using Silvaco's powerful toolset.

Refer: [Atlas Silvaco Official Manual](#) for detailed user Guide

Youtube Tutorial Suggestion: [click here](#)

Prerequisites:

- **Basic semiconductor knowledge:** understanding doping, P-N junctions, MOSFET operation, and fabrication steps.
- **Familiarity with Linux or Windows command line** for running simulations in DeckBuild, Silvaco's unified development environment. [Silvaco+1](#)
- **Silvaco TCAD software installed**, ideally the latest (2024 Baseline) version, with hardware requirements and OS compatibility checked. [Silvaco](#)

Targeted Audience:

- Graduate and undergraduate students (like M.Tech-level folks!) beginning their semiconductor device simulation journey.
- Newly inducted researchers or engineers transitioning from theory-heavy roles to computational device modeling.

Starting with process simulations in ATHENA and transitioning to device-level exploration in ATLAS, we'll include real-world script examples, mesh strategies, physics model explanations, and visualization tips using TonyPlot.

Chapter 1 – Introduction to TCAD and the Silvaco Environment

Technology Computer-Aided Design (TCAD) refers to the use of specialized software tools to model, simulate, and analyze semiconductor devices and processes before fabrication. By employing TCAD, engineers and researchers can optimize device performance, reduce prototyping costs, and predict electrical, thermal, and optical characteristics under various conditions—without physically producing wafers.

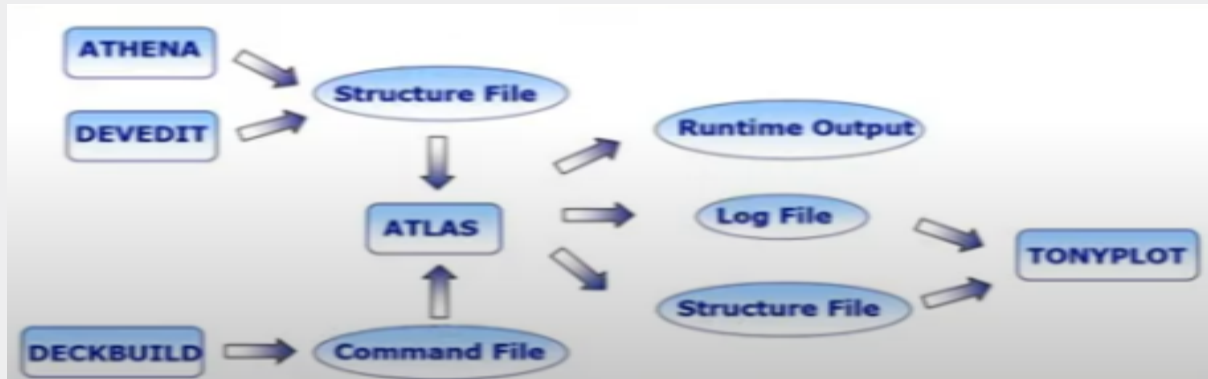
1.1 The Role of TCAD in Semiconductor Engineering

In the semiconductor industry, scaling challenges, material limitations, and process variability make device design increasingly complex. TCAD acts as a virtual laboratory, allowing controlled experimentation with device geometries, doping profiles, and fabrication steps. Through simulation, users can explore the impact of design changes on key parameters such as threshold voltage, leakage current, breakdown voltage, and switching speed.

1.2 Silvaco TCAD Suite Overview

Silvaco's TCAD suite is composed of several integrated tools:

- **ATHENA** - Process simulation to replicate fabrication steps, such as oxidation, implantation, etching, and diffusion.
- **ATLAS** - Device simulation that solves semiconductor equations to predict electrical and optical behavior.
- **DevEdit** - Structure creation and modification tool for both process and device simulations.
- **TonyPlot** - Visualization software for examining simulation results, including doping profiles, band diagrams, and I-V curves.



1.3 Benefits of Using Silvaco TCAD

- **Cost Reduction** - Eliminates multiple fabrication iterations.
- **Speed** - Enables rapid design modifications.
- **Flexibility** - Supports diverse devices, from MOSFETs and BJTs to LEDs, photodiodes, and emerging nanodevices.

1.4 How the Simulation Workflow Operates

A typical Silvaco TCAD workflow follows these steps:

1. Define the device geometry and mesh using ATHENA or DevEdit.
2. Assign material properties, doping profiles, and process steps.
3. Configure electrodes and bias conditions in ATLAS.
4. Select appropriate physics models for carrier transport, recombination, and mobility.
5. Run the simulation and analyze results using TonyPlot.

1.5 Beginner Focus of This Tutorial

This documentation targets first-time users—particularly students and junior engineers—who want a practical, hands-on entry point to TCAD. Each chapter combines theoretical background with executable examples, mirroring the flow demonstrated in the referenced Silvaco TCAD video tutorials. The learning curve is kept gradual, starting with simple 2D structures before moving to advanced simulations such as short-channel MOSFETs, BJTs, and photonic devices.

By the conclusion of this tutorial, readers will be able to:

- Create and edit device structures.
- Implement doping and material configurations.
- Apply relevant semiconductor physics models.
- Generate and interpret key plots for device characterization.

Chapter 2 – Silvaco TCAD Environment Familiarization

Silvaco TCAD's simulation environment consists of several integrated tools working together to perform process and device simulations efficiently. Understanding these tools and their file interactions is crucial for a smooth workflow.

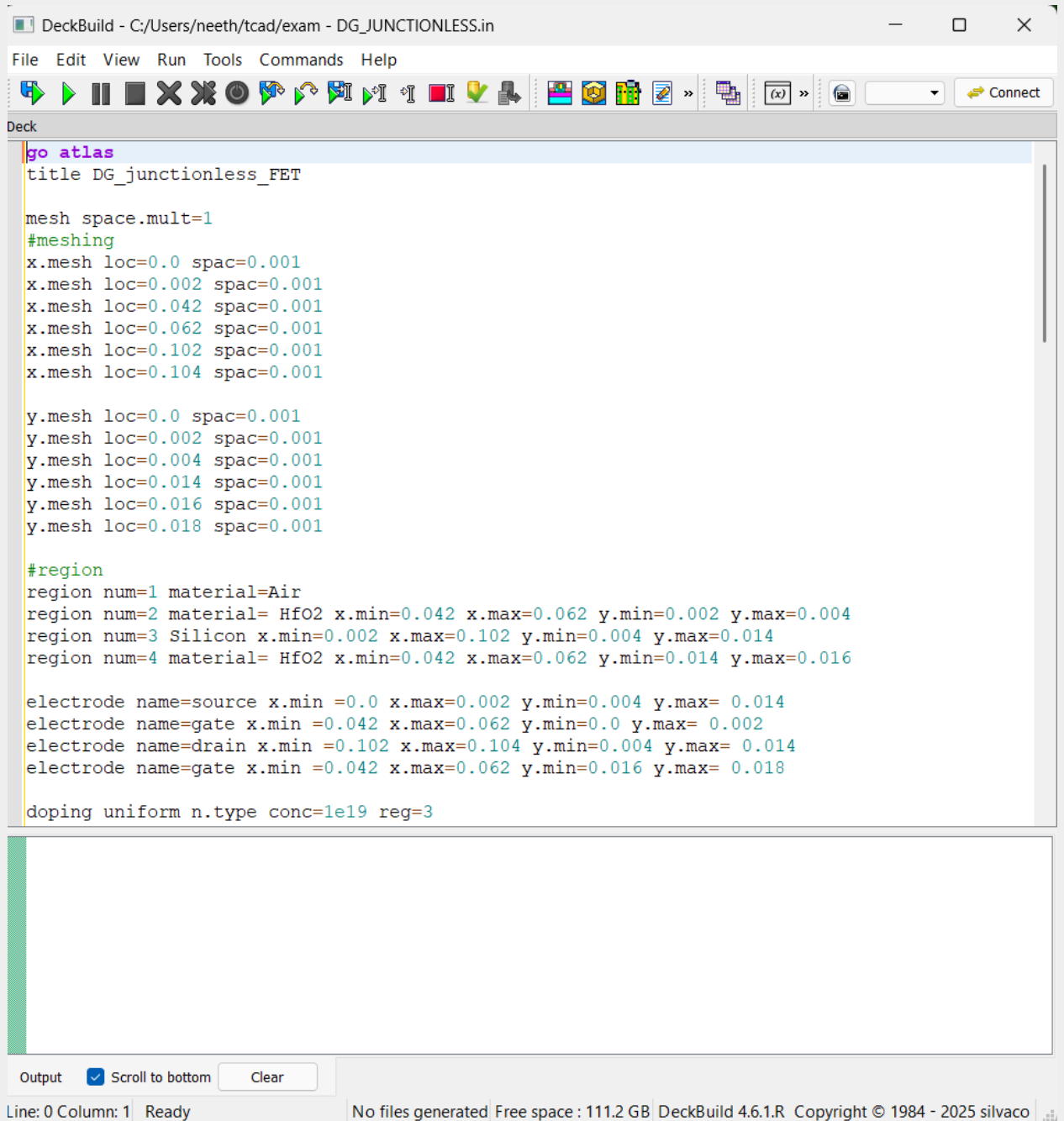
2.1 DeckBuild – The Simulation Control Center

DeckBuild is the main interface for creating, running, and debugging simulation scripts. It supports both ATHENA and ATLAS commands within a single environment.

Key Features:

- **Editor Window** – Write or load `.in` input scripts.
- **Control Buttons** – Run, stop, or step through simulations.

- **Output Window** - Displays real-time messages, warnings, and errors.



The screenshot shows the DeckBuild application window. The title bar reads "DeckBuild - C:/Users/neeth/tcad/exam - DG_JUNCTIONLESS.in". The menu bar includes "File", "Edit", "View", "Run", "Tools", "Commands", and "Help". The toolbar contains various icons for file operations, simulation control, and visualization. The main area is a code editor with the following text:

```

go atlas
title DG_junctionless_FET

mesh space.mult=1
#meshing
x.mesh loc=0.0 spac=0.001
x.mesh loc=0.002 spac=0.001
x.mesh loc=0.042 spac=0.001
x.mesh loc=0.062 spac=0.001
x.mesh loc=0.102 spac=0.001
x.mesh loc=0.104 spac=0.001

y.mesh loc=0.0 spac=0.001
y.mesh loc=0.002 spac=0.001
y.mesh loc=0.004 spac=0.001
y.mesh loc=0.014 spac=0.001
y.mesh loc=0.016 spac=0.001
y.mesh loc=0.018 spac=0.001

#region
region num=1 material=Air
region num=2 material= HfO2 x.min=0.042 x.max=0.062 y.min=0.002 y.max=0.004
region num=3 Silicon x.min=0.002 x.max=0.102 y.min=0.004 y.max=0.014
region num=4 material= HfO2 x.min=0.042 x.max=0.062 y.min=0.014 y.max=0.016

electrode name=source x.min =0.0 x.max=0.002 y.min=0.004 y.max= 0.014
electrode name=gate x.min =0.042 x.max=0.062 y.min=0.0 y.max= 0.002
electrode name=drain x.min =0.102 x.max=0.104 y.min=0.004 y.max= 0.014
electrode name=gate x.min =0.042 x.max=0.062 y.min=0.016 y.max= 0.018

doping uniform n.type conc=1e19 reg=3
  
```

Below the code editor is an empty output window. At the bottom of the application, there is a status bar with the following information:

Line: 0 Column: 1 Ready | No files generated | Free space : 111.2 GB | DeckBuild 4.6.1.R Copyright © 1984 - 2025 silvaco

2.2 Loading Example Files

Silvaco provides pre-built example scripts to accelerate learning.

To load an example:

1. In DeckBuild, go to **File** → **Open Example**.
 2. Select a relevant category (e.g., MOSFET, BJT, Diode).
 3. Review the script and modify parameters before running.
These examples serve as templates for your own projects.
-

2.3 TonyPlot - Visualization Tool

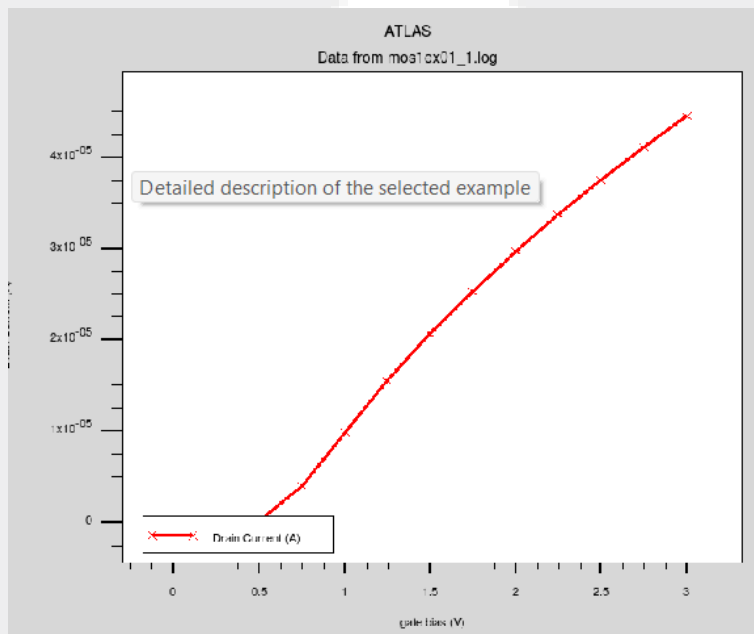
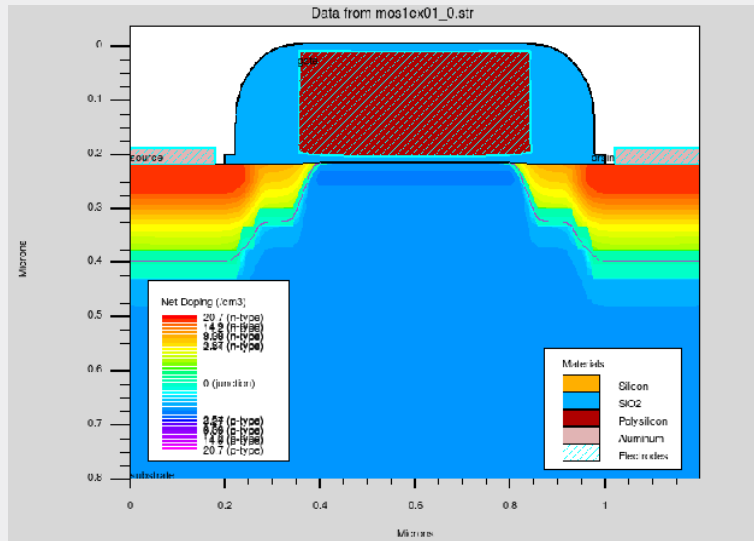
TonyPlot is used to view simulation results in graphical form.

Common uses:

- Plot doping concentration vs. depth.
- Display I-V or C-V curves.
- View band diagrams and electric field profiles.

Command-line usage:

```
bash
CopyEdit
tonyplot filename.plt
tonyplot filename.str
```



2.4 Log Files (.log)

Generated after a simulation run, log files store:

- Simulation parameters.
- Model equations used.
- Output results such as threshold voltage, current values, or breakdown voltages.

Always review log files for warnings or errors before analyzing results.

2.5 Structure Files (.str)

Structure files contain the geometry, mesh, and doping profile of the device after a process or device simulation step.

- Can be loaded in **DevEdit** for modification.
- Can be re-used in ATLAS simulations without re-running ATHENA.

Chapter 3 – Getting Started with DeckBuild

DeckBuild is the central simulation interface in the Silvaco TCAD suite, acting as a unified platform for writing, editing, running, and debugging simulation scripts. It supports both ATHENA (process simulation) and ATLAS (device simulation) commands, allowing seamless workflow management from structure creation to result analysis.

2.1 Launching DeckBuild

After installing Silvaco TCAD, ensure the environment variables are properly set. Launch DeckBuild via:

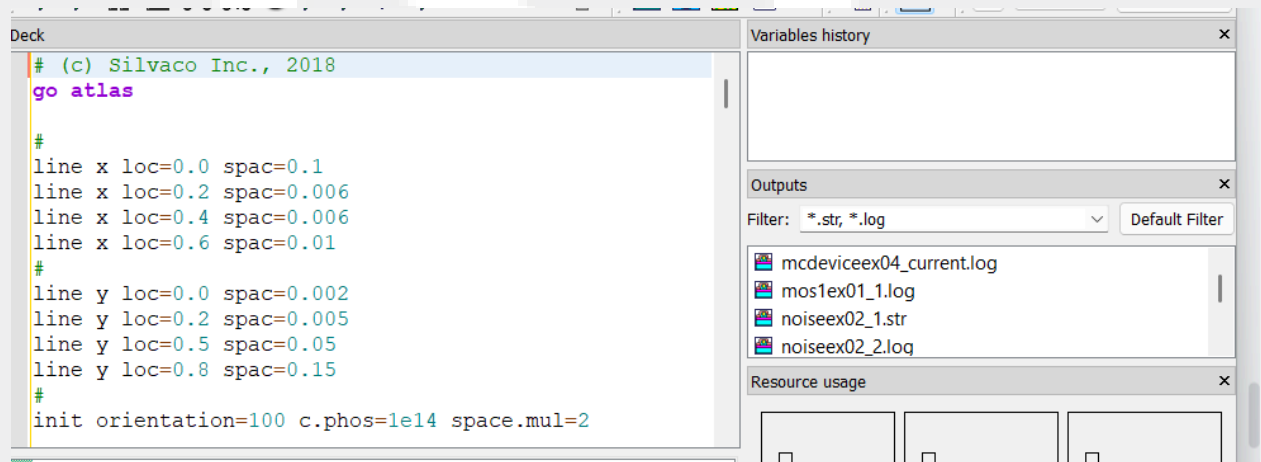
- **Windows** – Use the desktop shortcut or run `deckbuild` from the Silvaco command prompt.

Linux – Open a terminal in your project directory and enter:

```
deckbuild &
```

-

It is recommended to always work in a dedicated project folder to keep `.in`, `.log`, `.str`, and `.plt` files organized.



2.3 Supported File Types

- `.in` – Input script containing simulation commands.
- `.log` – Text log file with messages and results.

- `.str` - Structure file storing device geometry and doping profiles.
- `.plt` - Plot file for TonyPlot visualization.

2.4 Writing Your First Script

A simple ATLAS start command looks like:

```
go atlas
mesh space.mult=1
...
```

Scripts can be typed directly or loaded from existing examples.

2.5 Running and Debugging

- Click **Run** to execute the loaded script.
- Monitor the **Output Window** for progress.
- If errors occur, DeckBuild highlights the line number for correction.

2.6 Best Practices for Beginners

- Keep scripts modular (separate mesh, doping, and simulation sections).
- Use comments (`!` in scripts) for readability.
- Save and version your scripts to track changes.

With DeckBuild mastered, you are ready to start defining your first process or device simulation in Silvaco TCAD. The next chapter will focus on **creating and meshing device structures**—the foundation of accurate simulations.

Chapter 4 – PN Junction Diode Simulation

Aim

To simulate and analyze the electrical characteristics of a PN junction diode using Silvaco TCAD, and extend the methodology to an NPN BJT structure for comparative understanding.

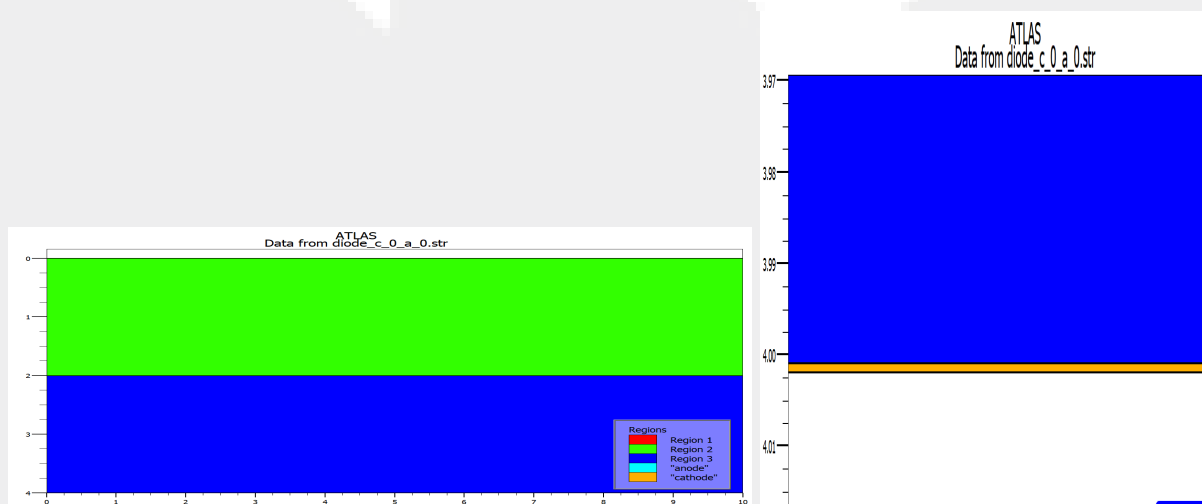
Problem Statement

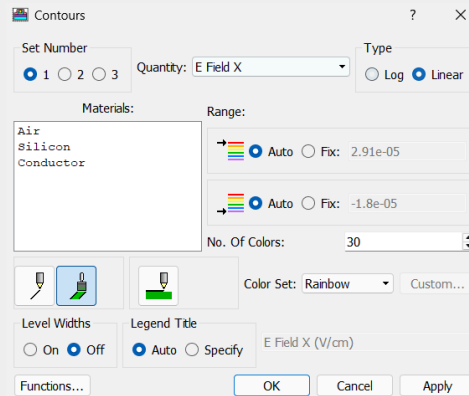
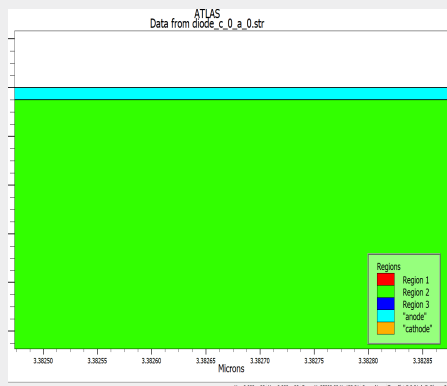
Design a semiconductor structure in Silvaco ATLAS to observe the electrical behavior of a PN junction under forward and reverse bias. Define the geometry, doping concentrations, and mesh for accurate simulation results. Then, extend the concept to a three-region NPN BJT, implementing correct doping levels for the collector, base, and emitter regions. The objective is to generate I-V characteristics and examine the influence of doping profiles on device performance.

ATLAS Script –

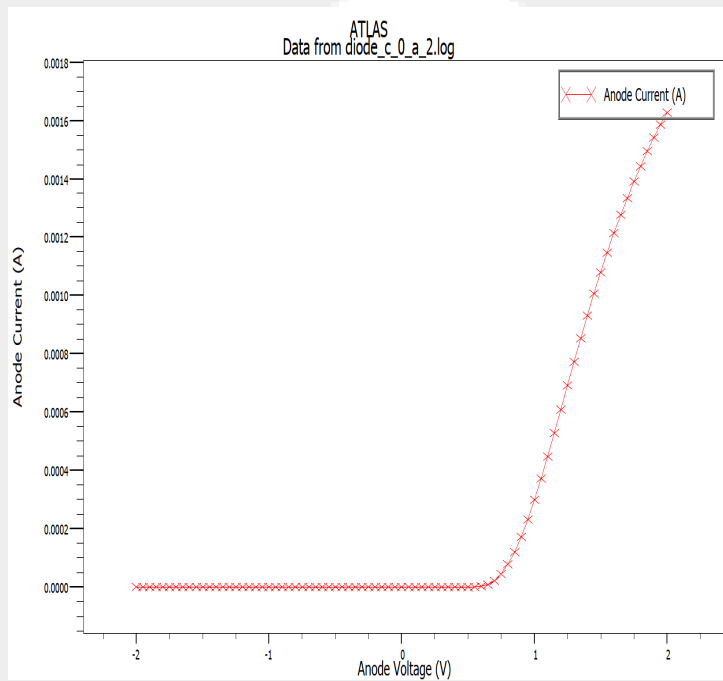
https://github.com/Neethu-Jaisan/Silvaco_TCAD/blob/main/pndiode.in

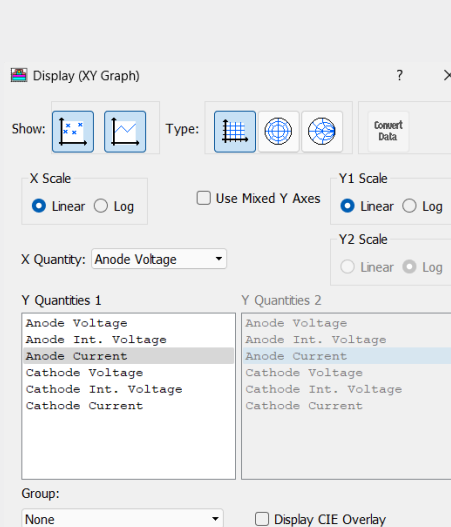
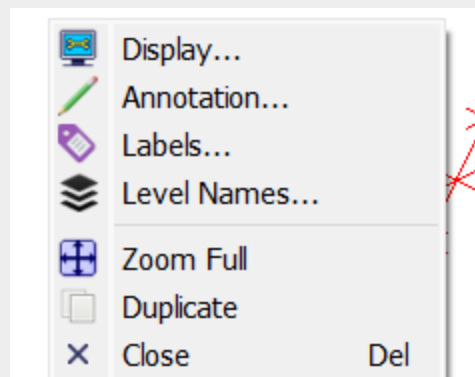
Tony Plot





Using contours from display you can view different physical quantities and their nature in the diode.





Here you can

change parameters for the graph

Chapter 5 - Examples for you to Do

Bipolar Junction Transistor

https://github.com/Neethu-Jaisan/Silvaco_TCAD/blob/main/bjt.in

Aim:

To design and simulate an NPN Bipolar Junction Transistor (BJT) in Silvaco TCAD ATLAS by defining the geometry, doping profiles, and bias conditions, and to analyze its I-V characteristics through incremental base voltage sweeps.

Device Specifications:

- **Overall Structure:** Vertical NPN BJT
 - **Collector Region:** Lightly doped n-type silicon ($1 \times 10^{16} \text{ cm}^{-3}$)
 - **Base Region:** Moderately doped p-type silicon ($1 \times 10^{17} \text{ cm}^{-3}$)
 - **Emitter Region:** Heavily doped n-type silicon ($1 \times 10^{19} \text{ cm}^{-3}$)
 - **Total Device Width:** 12.0 μm (X-direction)
 - **Total Device Height:** 6.5 μm (Y-direction)
 - **Base Thickness:** 3.5 μm
 - **Emitter Height:** 3.0 μm
-

Simulation Requirements:

1. Mesh Design:

- a. Fine mesh in emitter-base junction region for accurate electric field and carrier concentration capture.

- b. Coarser mesh outside active junction areas to optimize computation time.

2. Regions:

- a. **Region 1:** Collector (n-type, light doping)
- b. **Region 2:** Base (p-type, moderate doping)
- c. **Region 3:** Emitter (n-type, heavy doping)

3. Doping:

- a. Collector: n-type, $1 \times 10^{16} \text{ cm}^{-3}$
- b. Base: p-type, $1 \times 10^{17} \text{ cm}^{-3}$
- c. Emitter: n-type, $1 \times 10^{19} \text{ cm}^{-3}$

4. Electrodes:

- a. **Collector:** Top contact
- b. **Base:** Bottom contact under base region
- c. **Emitter:** Bottom contact under emitter region

5. Models Used:

- a. **SRH** (Shockley-Read-Hall recombination)
- b. **FLDMOB** (field-dependent mobility)
- c. **BGN** (bandgap narrowing)

6. Biasing & Analysis:

- a. Collector voltage fixed at 1 V
- b. Base voltage swept from 0.3 V to 0.9 V in 0.05 V steps
- c. Output: Collector current (I_C) vs. Base voltage (V_B)

MOSFET Simulation

https://github.com/Neethu-Jaisan/Silvaco_TCAD/blob/main/MOSFET.in

Aim:

To design, simulate, and analyze the electrical characteristics of an n-channel MOSFET using Silvaco TCAD ATLAS. The simulation will define the MOSFET geometry, material regions, doping concentrations, and electrode placements to generate I-V characteristics for specified gate voltages.

Device Specifications:

- **Channel Length (L):** 0.05 μm
 - **Gate Oxide Thickness (t_{ox}):** 0.002 μm (2 nm)
 - **Source and Drain Regions:** Highly doped n-type silicon
 - **Substrate:** P-type silicon with high doping concentration
 - **Air Region:** Above gate oxide for simulation environment boundary
 - **Width:** 0.25 μm total simulation domain in X-direction
 - **Height:** 0.016 μm total simulation domain in Y-direction
-

Simulation Requirements:

1. Mesh Definition:

- Fine mesh in critical regions (gate oxide, channel) for higher accuracy.
- Coarser mesh in non-critical areas to optimize runtime.

2. Regions:

- Region 1: Air (for environment boundary)
- Region 2: Gate oxide (SiO_2)
- Regions 3, 5: Source and drain silicon
- Region 4: Channel silicon

3. Doping:

- Source/Drain: n-type, $1 \times 10^{18} \text{ cm}^{-3}$
- Channel: p-type, $1 \times 10^{20} \text{ cm}^{-3}$
- Gate oxide: n-type, $1 \times 10^{18} \text{ cm}^{-3}$ (as per given script)

4. Electrodes:

- **Drain:** Left silicon contact
- **Gate:** Above oxide region
- **Source:** Right silicon contact

5. Models:

- **bipolar, fldmob** (field-dependent mobility model)

6. Biasing & Analysis:

- Fixed $V_G = 0.5 \text{ V}$
- V_D swept from 0 V to 3 V in 0.1 V steps
- Output: Drain current (I_D) vs. Drain voltage (V_D)

SOI MOSFET Simulation

https://github.com/Neethu-Jaisan/Silvaco_TCAD/blob/main/Soi_Mosfet.in

Aim

- To design, simulate, and analyze the electrical characteristics of an n-channel SOI MOSFET using Silvaco TCAD ATLAS. The simulation will define the device geometry, material layers, doping profiles, and electrode placements to generate I-V characteristics for specified gate and drain voltages.

Device Specifications

- Channel Length (L): 0.05 μm
- Gate Oxide Thickness (t_{ox}): 0.002 μm (2 nm)
- Silicon Film Thickness (t_{si}): 0.01 μm (10 nm)
- Buried Oxide (BOX) Thickness: 0.02 μm (20 nm)
- Source and Drain Regions: Highly doped n⁺ silicon
- Body Region: Lightly doped p-type silicon
- Air Region: Above gate oxide for environment boundary
- Width: 0.25 μm (simulation domain in X-direction)
- Height: 0.05 μm (total domain in Y-direction)

Simulation Requirements

- Mesh Definition
- Fine mesh in channel, gate oxide, and silicon film regions for higher accuracy.
- Coarser mesh in air and BOX regions to reduce runtime.

Regions

- Air: Upper boundary for environment
- Gate Oxide (SiO_2): Thin insulating layer
- Silicon Channel: Lightly doped p-type
- Source/Drain: Heavily doped n^+ regions
- Buried Oxide (BOX): Thick SiO_2 layer
- Substrate: Mechanical support (ignored electrically, floating boundary)

Doping

- Source/Drain: n-type, $1 \times 10^{18} \text{ cm}^{-3}$
- Channel (Si film): p-type, $1 \times 10^{17} \text{ cm}^{-3}$
- BOX & Air: Intrinsic/insulating

Electrodes

- Gate: On top of gate oxide
- Source: Left silicon contact
- Drain: Right silicon contact

Models

- bipolar - Impact ionization
- fldmob - Field-dependent mobility
- cvt - Concentration-dependent mobility & velocity saturation
- srh - Shockley-Read-Hall recombination

Biasing & Analysis

- Transfer Characteristics (I_d - V_g):
- $V_d = 0.05$ V (linear region) and 1 V (saturation)
- V_g swept from 0-1.2 V in 0.02 V steps
- Output Characteristics (I_d - V_d):
- V_g fixed at 0.5 V, 0.7 V, 1.0 V
- V_d swept from 0-1 V in 0.05 V steps

Outputs

- I_d - V_g curves: Threshold voltage (V_{th}), subthreshold slope (SS), I_{on}/I_{off} ratio
- I_d - V_d curves: Channel length modulation, drive current capability

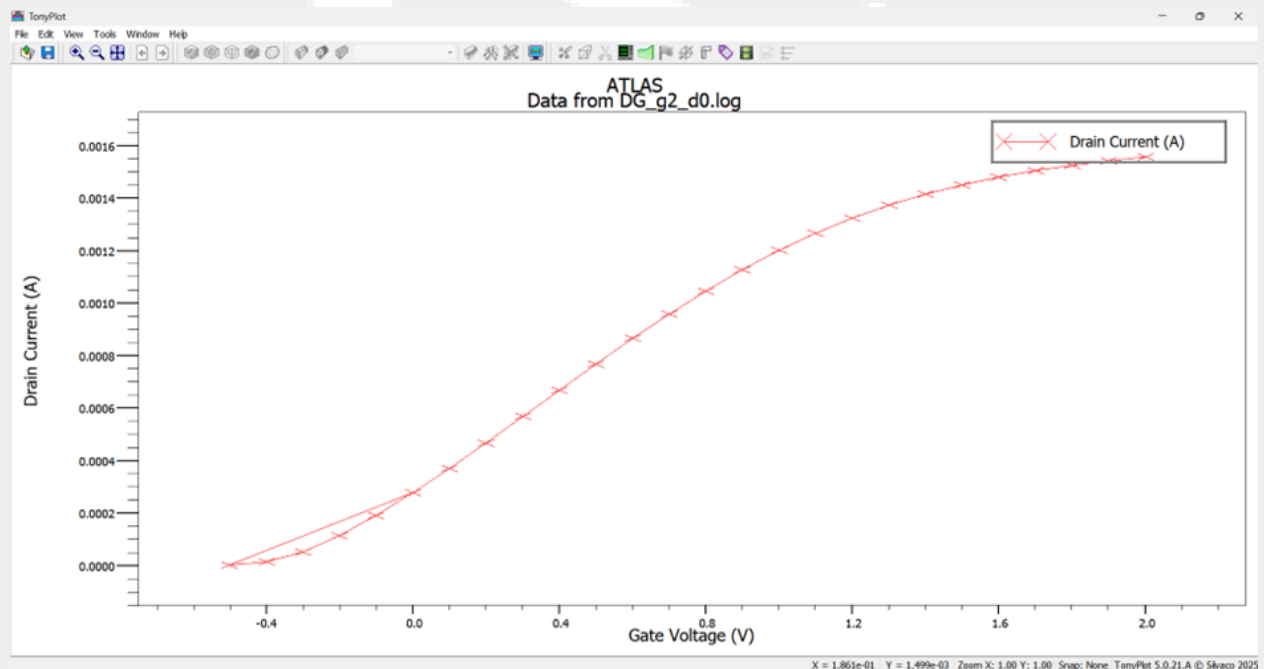
SOI Junctionless MOSFET Simulation

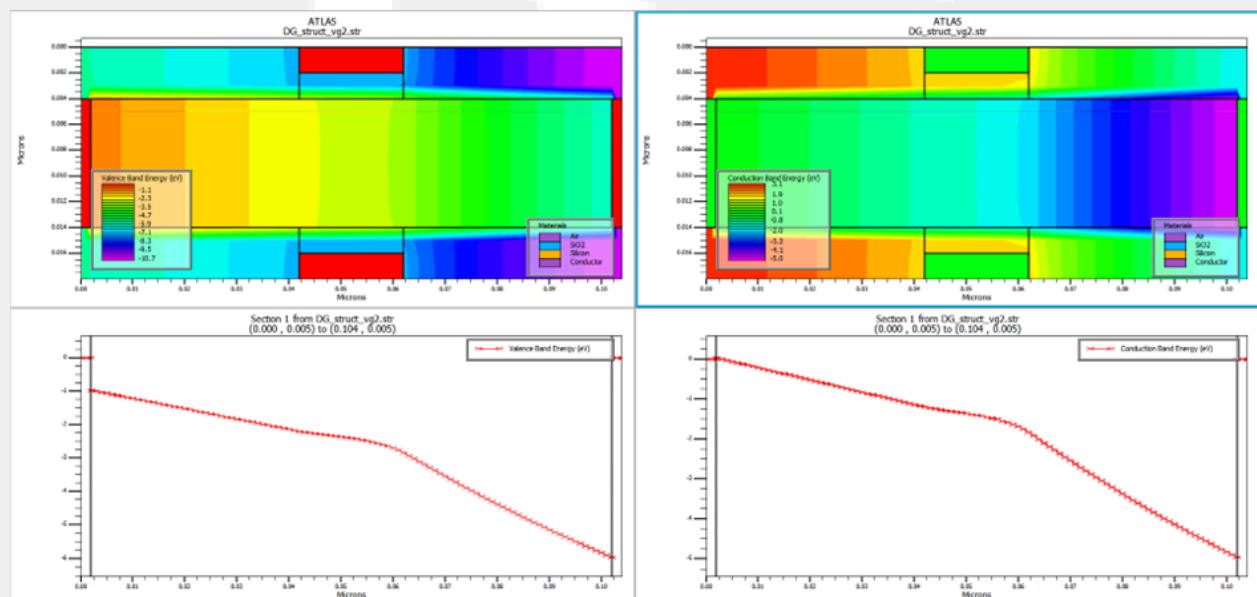
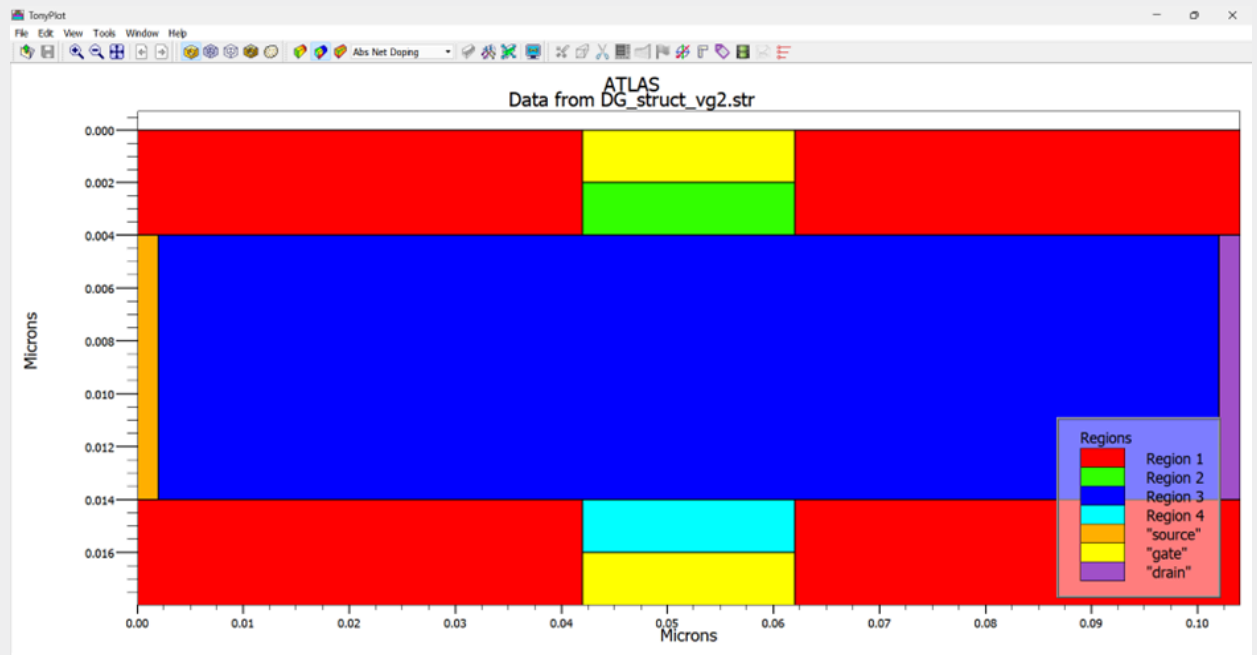
https://github.com/Neethu-Jaisan/Silvaco_TCAD/blob/main/soi_junctionless.in

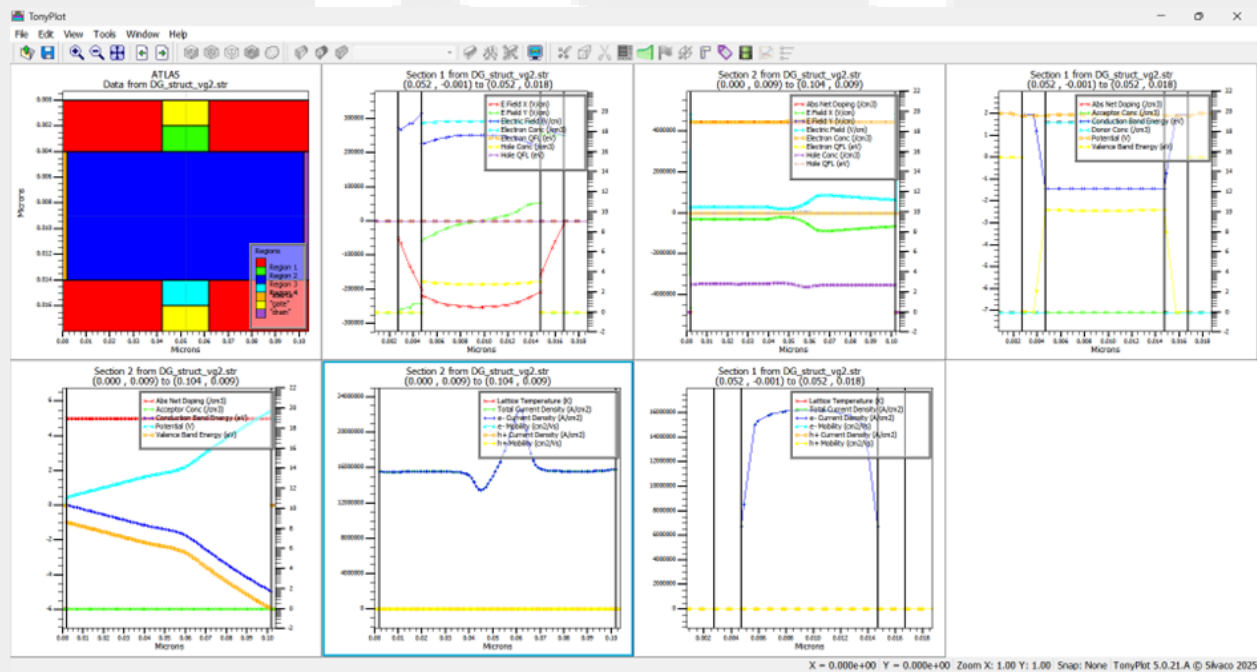
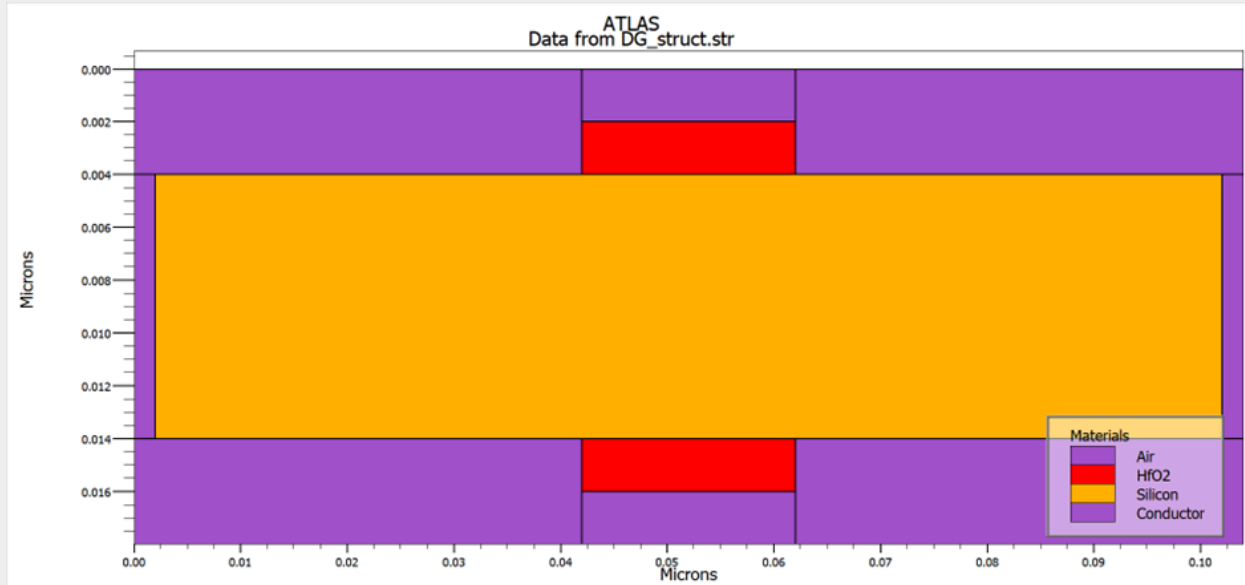
Aim:

To design, simulate, and analyze the electrical characteristics of a Junctionless SOI MOSFET using Silvaco TCAD ATLAS. The project focuses on defining device geometry, material regions, uniform channel doping, and electrode placement to evaluate short-channel effects and device performance. The simulation will generate I-V characteristics, extract key parameters such as threshold voltage (V_t), subthreshold slope (SS), drain-induced barrier lowering (DIBL), I_{on}/I_{off} ratio, and transconductance (g_m).

Simulation Results








```

ATLAS>
EXTRACT> init inf="DG_20_Vd_1.log"
EXTRACT> extract name="vt2" x.val from curve ((v."gate"),log10(abs(i."drain")))) where y.val=-7
vt2=0.705812
EXTRACT>
EXTRACT> #tonyplot DG_Ch_20_Vg_2_Vd_1fix.log
EXTRACT>
EXTRACT> extract name="dibl" (0.705163-0.705812)/(1-0.05)
dibl=-0.000683158
EXTRACT> #extract init infile="DG_d_5.log"
EXTRACT> extract name="subvt" 1.0/slope(maxslope(curve(v."gate",log10(abs(i."drain")))))
subvt=0.0734593 V/decade
EXTRACT> extract name="Ioff_n" y.val from curve(v."gate",i."drain") where x.val=0
Ioff_n=9.89652e-005
EXTRACT> extract name="Ion_n" y.val from curve(v."gate",i."drain") where x.val=1
Ion_n=0.00101107
EXTRACT> extract name="Ion_Ioff" 0.00101107/9.89652e-005
Ion_Ioff=10.2164
EXTRACT> extract name="vt" x.val from curve ((v."gate"),log10(abs(i."drain")))) where y.val=-7
vt=0.705812 V
EXTRACT>
EXTRACT> tonyplot -overlay DG_20_Vd_005.log DG_20_Vd_1.log -set DG_20_Vd_1.log
EXTRACT>
EXTRACT> quit
quit
ATLAS version 5.26.1.R finished at Fri Aug 29 11:52:51 2025

```

Short Channel Effect Parameters extracted and displayed in command window

Chapter 5 Summary and Questions

Summary

In this project, the design and simulation of semiconductor devices were carried out using **Silvaco TCAD ATLAS**. Starting from the basics of a **PN diode**, the work progressed through **BJT**, **n-channel MOSFET**, **SOI MOSFET**, and finally a **Junctionless SOI MOSFET**.

- The **PN diode** simulation helped understand forward/reverse bias behavior and I-V characteristics.
- The **BJT** experiment demonstrated current gain (β), input/output characteristics, and transistor action.
- The **MOSFET** section analyzed scaling, Id-Vd curves, and gate control.
- The **SOI MOSFET** introduced body isolation and reduction of parasitic capacitances.
- The **Junctionless SOI MOSFET** explored advanced short-channel effects, extracting critical metrics like **Vt**, **DIBL**, **Subthreshold Slope (SS)**, **Ion/Ioff ratio**, and **gm**.

Questions

Objective Type

1. Define the term **threshold voltage (V_t)** in MOSFETs.
2. What is **DIBL** and why does it occur?
3. State the advantage of **SOI MOSFETs** over bulk MOSFETs.
4. What is the typical range of **subthreshold slope (SS)** in MOSFETs?
5. Why is **uniform doping** used in a junctionless MOSFET?
6. In a PN diode, what causes the **built-in potential**?
7. Which mobility models were enabled in the simulation?

Short Descriptive Type

1. Explain the working principle of a **Junctionless MOSFET**.
2. Discuss the role of **gate oxide thickness** in MOSFET performance.
3. Compare **BJT and MOSFET** with respect to current control.
4. How does **SOI technology** help in reducing leakage currents?
5. What is the significance of the **I_{on}/I_{off} ratio**?
6. Derive the condition for the saturation **region** in MOSFET operation.
7. Explain how **Newton's method** is used for numerical convergence in TCAD.

References

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7. <https://youtu.be/ZY0bU1OMiKc>
8. <https://youtu.be/TOOsGR51TJE>
9. https://github.com/Neethu-Jaisan/Silvaco_TCAD
10. <https://www.silvaco.com>